

A toy model of deep learning

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Motivation 1: Implicit biases of the training process have safety implication

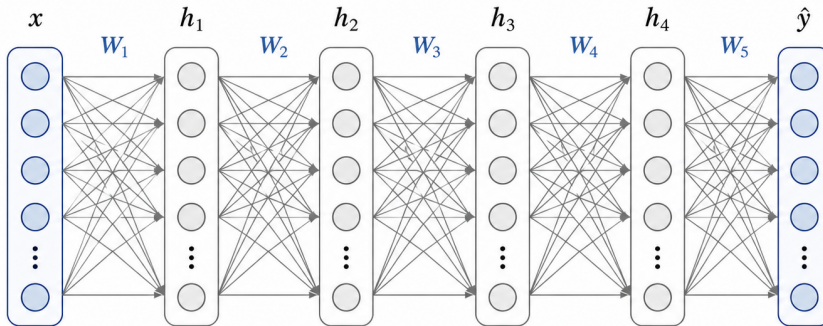
- Parameter $\theta \in \Omega$, input-output data $(Y, X) \sim P$, Neural network $f(\theta, x)$
- Degenerate functions: on input x many parameters are compatible with low loss $f(\theta, x) \approx f(\theta', x)$
- Indistinguishable on training and eval data
- However for some out of distribution input x_o , model $f(\theta, x_o)$ is unsafe and $f(\theta', x_o)$ is safe
- The data, hyperparameters, optimization determine if training favour safe or unsafe model

Motivation 2: Interpretability over training

- Challenging to know what are principled observables to track
- Learning dynamics can help defining useful observables
- Challenging to reverse engineer a neural network after training
- Structure often emerges sequentially across different timescales
- Might be easier to decompose the internal structure of a model during training

Setup

Deep Linear Network



All transformations are linear. Each layer performs a linear map: $h_k = W_k h_{k-1} \quad (k = 1, \dots, 5)$

$$\hat{y} = W_5 W_4 W_3 W_2 W_1 x$$

The harmonic oscillator of deep learning

Definition

A L -layer Deep Linear Network is a function:

$$f : \mathbb{R}^{d_0} \times \prod_{l=1}^{L-1} \text{Mat}_{d_l \times d_{l-1}} \rightarrow \mathbb{R}^{d_L}$$
$$(x, \theta := (W_1, \dots, W_L)) \mapsto W_L \dots W_1 x = Wx$$

d_0 : dimension of the input data; d_l , $1 \leq l \leq L$: width of layer l student matrix W .

DLN: multi-linear function, linear in W_l and in the input data x and linear in W .

The product map

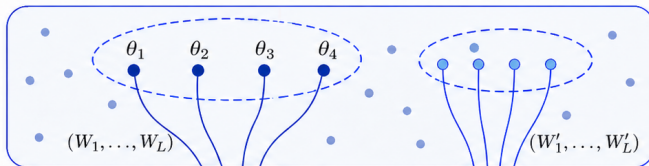
A depth- L deep linear network satisfies $f(\theta, x) = \mu(\theta)x$

Product map: $\mu : \Omega \rightarrow \mathcal{F}$

$$\theta = (W_1, \dots, W_L) \mapsto W_L \cdots W_1 = W$$

Parameter space Ω

micro / upstairs



fiber $\mu^{-1}(W)$

many parameters,
same map

fiber $\mu^{-1}(W')$

many parameters,
same map

Function space $\mathcal{F}$

macro / downstairs



fiber $\mu^{-1}(W')$

(another fiber)

A DLN is **linear** in function space $\mathcal{F}$ but **nonlinear** in parameter space Ω .

Linear over **function space $\mathcal{F}$**

Nonlinear over **parameter space Ω**



Degenerate solutions: a fiber can be a whole algebraic variety.

The product map

Definition

A DLN $f(\theta, x)$ of depth L satisfies $f(\theta, x) = \mu(\theta)x$ with *product map* μ :

$$\begin{aligned}\pi : \Omega = \text{Mat}_{d_I \times d_{I-1}} &\rightarrow \mathcal{F} = \text{Mat}_{d_0 \times d_L} \\ \theta := (W_1, \dots, W_L) &\mapsto W_L \dots W_1 = W\end{aligned}$$

We note $W := \pi(\theta)$. A DLN is linear over function space $\mathcal{F}$ and non-linear over parameter space Ω .

Intuition: Parameter space Ω is micro (upstair) and $\mathcal{F}$ is macro (downstair).

Degenerate solutions: fiber $\mu^{-1}(\theta)$ can be a whole algebraic variety.

Quadratic Population loss and gradient flow

Teacher matrix M such that $y = Mx$ for x generated by $N(0, I)$ distribution. Input-output covariance $\Sigma_{YX} = M$

$$\mathcal{L}(\theta) := \frac{1}{2} \|(M - W_L \dots W_1)\|_F^2 = \frac{1}{2} \|(M - W)\|_F^2$$

Initialize $\theta_0 \sim N(0, \sigma^{-\gamma})$ with γ init. scale, then gradient flow:

$$\dot{W}_l = -\nabla_{W_l} L(\theta)$$

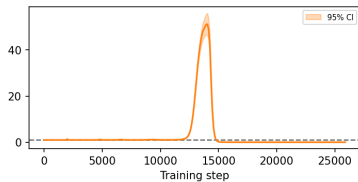
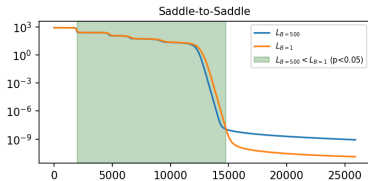
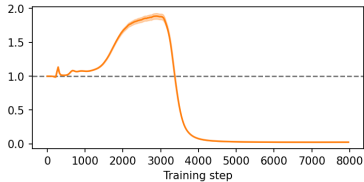
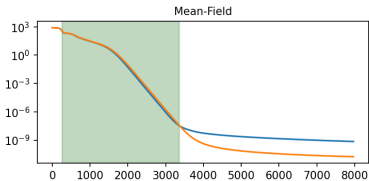
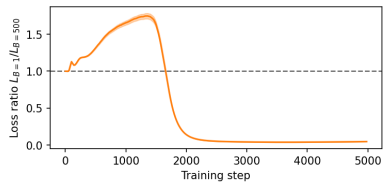
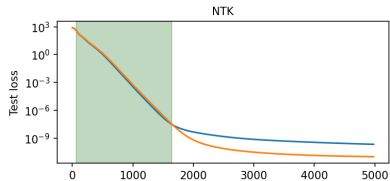
Frobenius norm as $\|A\|_F := \sqrt{\text{Tr}(A^\top A)}$

Global minimizer achieving zero loss: $\exists \theta^* \in \Omega, \mu(\theta^*) = W^* = M$

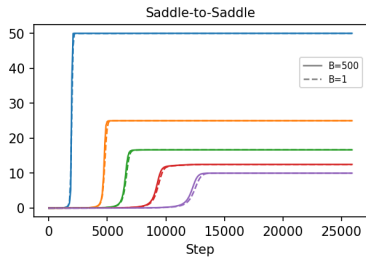
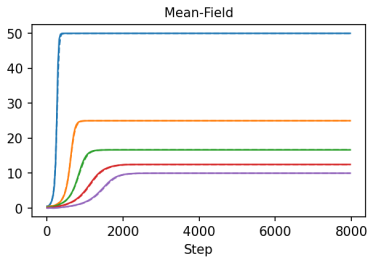
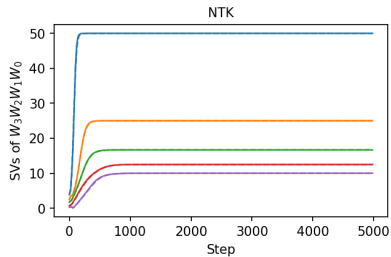
Structure of the data encoded by Σ_{YX}

Empirical phenomena

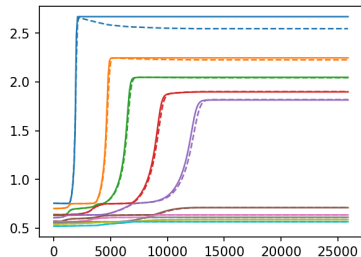
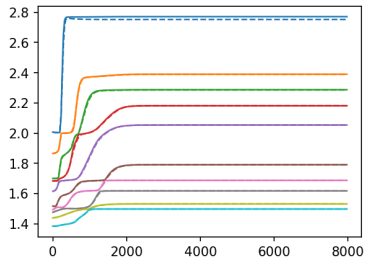
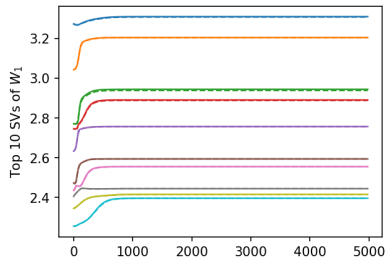
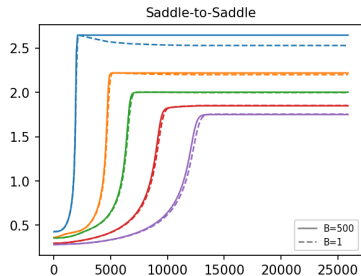
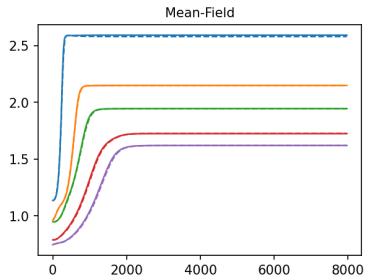
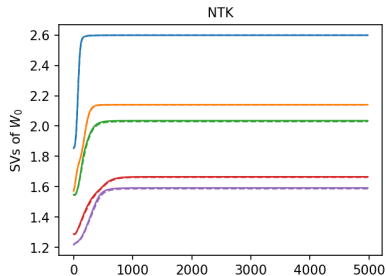
Training losses



Singular values of student matrix



Singular values of individual weight matrices



Geometry

A 2-layer diagonal DLN

Model and loss

Take a 2-layer diagonal DLN of width 2:

$$W_1 = \text{diag}(b_1, b_2), \quad W_2 = \text{diag}(a_1, a_2), \quad \theta = (W_1, W_2).$$

Hence the student matrix is

$$W = W_2 W_1 = \text{diag}(a_1 b_1, a_2 b_2).$$

Let the teacher be

$$M = \text{diag}(s_1, s_2), \quad s_1, s_2 > 0, \quad x \sim \mathcal{N}(0, I).$$

With whitened inputs, the quadratic population loss is

$$\mathcal{L}(\theta) = \frac{1}{2} \|M - W\|_F^2 = \frac{1}{2} (a_1 b_1 - s_1)^2 + \frac{1}{2} (a_2 b_2 - s_2)^2.$$

Gradient equations

For $i = 1, 2$,

$$\partial_{a_i} \mathcal{L} = (a_i b_i - s_i) b_i, \quad \partial_{b_i} \mathcal{L} = (a_i b_i - s_i) a_i.$$

Therefore the critical point equations are

$$(a_i b_i - s_i) b_i = 0, \quad (a_i b_i - s_i) a_i = 0, \quad i = 1, 2.$$

The two coordinates decouple: each singular direction can be analyzed independently.

Coordinate-wise characterization

For each $i = 1, 2$, the critical point equations are

$$(a_i b_i - s_i) b_i = 0, \quad (a_i b_i - s_i) a_i = 0.$$

Since $s_i > 0$, this implies exactly two possibilities:

$$a_i b_i = s_i \quad \text{or} \quad a_i = b_i = 0.$$

Thus each coordinate either fits the teacher singular value or is shut off.

Critical points

All critical points

Parameter conditions	Realized student matrix W	Loss
$a_1 b_1 = s_1, a_2 b_2 = s_2$	$W = \text{diag}(s_1, s_2) = M$	$\mathcal{L} = 0$
$a_1 b_1 = s_1, a_2 = b_2 = 0$	$W = \text{diag}(s_1, 0)$	$\mathcal{L} = \frac{1}{2} s_2^2$
$a_1 = b_1 = 0, a_2 b_2 = s_2$	$W = \text{diag}(0, s_2)$	$\mathcal{L} = \frac{1}{2} s_1^2$
$a_1 = b_1 = 0, a_2 = b_2 = 0$	$W = 0$	$\mathcal{L} = \frac{1}{2}(s_1^2 + s_2^2)$

Critical points correspond to choosing a subset of the teacher singular values.
Depth introduces three new critical points compared with the one layer DLN loss

$$L(W) = \|M - \text{diag}(w_1, w_2)\|_F^2$$

Example

If both singular directions are fitted,

$$a_1 b_1 = s_1, \quad a_2 b_2 = s_2,$$

then

$$W = M = \text{diag}(s_1, s_2), \quad \mathcal{L} = 0.$$

Since $\mathcal{L}(\theta) \geq 0$, these critical points are global minima.

Example: $W = \text{diag}(s_1, 0)$ is a saddle

Take a critical point with $a_1 b_1 = s_1$ and $a_2 = b_2 = 0$.

For the perturbation $a_2 = t$, $b_2 = t$,

$$W(t) = \text{diag}(s_1, t^2),$$

and

$$\mathcal{L}(t) - \mathcal{L}(0) = \frac{1}{2}(t^2 - s_2)^2 - \frac{1}{2}s_2^2 = -s_2 t^2 + \frac{1}{2}t^4 < 0$$

for small $t \neq 0$.

For the perturbation $a_2 = t$, $b_2 = -t$,

$$W(t) = \text{diag}(s_1, -t^2),$$

and

$$\mathcal{L}(t) - \mathcal{L}(0) = \frac{1}{2}(-t^2 - s_2)^2 - \frac{1}{2}s_2^2 = s_2 t^2 + \frac{1}{2}t^4 > 0.$$

Thus there are both descending and ascending directions, so this critical point is a saddle.

Deep learning without poor local minima

Assume that Σ_X and Σ_{YX} are full rank. Assume that Σ_{YX} has d_L distinct singular values.

No non-global local minima

The loss function of DLN of depth L is non-convex, non-concave, and every critical point that is not a global minimum is a saddle point.

The training dynamics cannot get stuck in spurious local minima.

Reference: *Kawaguchi, Kenji. "Deep learning without poor local minima." Advances in neural information processing systems 29 (2016)*

Parametrization of critical points

Teacher SVD: $M = U \operatorname{diag}(s_1, \dots, s_{r_m}, 0_{d_L - r_m}) V^\top$.

Saddle points

Let $\theta \in \Omega$ be a critical point of rank r . Then there exists a subset S of left singular vector of M and a projector $P_S = U_S U_S^\top$ over these singular vector such that:

$$\mu(\theta) = P_S M = U_S \operatorname{diag}(s_1, \dots, s_r, 0_{d_L - r}) V^\top$$

The saddle points are given by a subset of singular values and (left) singular vectors of the teacher M .

Reference: *Achour, El Mehdi, François Malgouyres, and Sébastien Gerchinovitz. "The loss landscape of deep linear neural networks: a second-order analysis." Journal of Machine Learning Research 25.242 (2024): 1-76.*

Symmetries

Let $G_{\mathbf{d}} := \prod_{l=L}^0 GL_{d_l} = GL_{d_L} \times GL_{d_h} \times GL_{d_0}$ acts on Ω via:

$$(W_1, \dots, W_L) \mapsto (g_1 W_1 g_0^{-1}, g_2 W_2 g_1^{-1}, \dots, g_L W_L g_{L-1}^{-1}),$$

Equivariance of the multiplication map

Since $W_L g_{L-1}^{-1} g_{L-1} W_{L-2} g_{L-2}^{-1} \dots g_1 W_1 = W_L \dots W_1$, the multiplication map μ is $GL_{\mathbf{d}}$ -equivariant and G_h -invariant i.e.:

$$\mu(g \cdot \theta) = g_L W g_0^{-1}; \quad \mu(g_h \cdot \theta) = \mu(\theta)$$

A DLN is degenerate and there are infinite DLNs that maps to a particular student.

Natural to study solutions up to $G_{\mathbf{d}}$ symmetries: by equivariance of μ we have a group action of $G_{\mathbf{d}}$ over the determinantal variety $G_{\text{out}} \cdot \mu^{-1}(\theta) = \Sigma_r^{\mathbf{d}} := \{\theta \in \Omega \mid \text{rank}(\mu(\theta)) = r\}$

Quiver representation theory

Basic notions

A quiver Q is a graph (I, E, s, t) with a map $s(e) = V_i$ and $t(e) = V_{i+1}$ when edge $e \in E$ connects vertices i and $i + 1$ in I . Loops and cycles are allowed.

A quiver representation assigns a vector space V_i to each $i \in I$ and a linear map ϕ_e to each arrow $e \in E$. Note $\text{Rep}_d(Q)$ the space of representations of dimension vector $d = (d_i)_{i \in I}$

Example: The linear quiver $A_n : 1 \rightarrow 2 \rightarrow \dots \rightarrow n$ and representation $k \xrightarrow{id} k \xrightarrow{id} \dots \xrightarrow{id} k$

DLN as quiver representations

The multiplication map of a DLN μ is a map from $\text{Rep}_d(A_{L+1}) \rightarrow \text{Mat}_{d_L \times d_0}$ with linear maps given by weight matrices W_i and vector space $\mathbb{R}^{d_i}$.

In particular Σ_d^r is a subvariety of $\text{Rep}_d(A_{L+1})$.

Classification of indecomposables of $\text{Rep}_d(A_{L+1})$

Indecomposable is smallest $M_i \in \text{Rep}_d(Q)$ such that $M \cong \bigoplus M_i^{m_i}$ for $M \in \text{Rep}_d(Q)$

Indecomposables of A_L

Example: $k^2 \rightarrow^{\phi_1} k^2 \rightarrow^{\phi_2} k \cong k \rightarrow^{id} k \rightarrow^0 0 \oplus k \rightarrow^{id} k \rightarrow^0 0$.

More generally all indecomposables are of the form:

$M(i, j) = 0 \rightarrow^0 \dots \rightarrow_i^0 k \rightarrow^{id} \dots \rightarrow^{id} k \rightarrow_{j+1}^0 0 \rightarrow \dots \rightarrow^0 0$ for $0 \leq i < j \leq L + 1$

See: Gabriel Theorem for Dynkin quivers.

Decomposition via rank patterns

Krull-Schmidt and Gabriel theorem $\implies$ for all $M \in \text{Rep}_d(A_{L+1})$ we have a unique decomposition:

$$M \cong \bigoplus_{r_{ij} \in \mathcal{R}_d} M(i, j)^{\oplus r_{ij}}$$

With $r_{ij} := \text{rank}(W_{i \rightarrow j})$ and $r_{ii} = d_i$ the *rank patterns*; and $\mathcal{R}_d$ the set of rank patterns (with some constraints)

Stratification of the detremimental variety

Orbits are parameterized by rank patterns

We have G_d action over Σ_d^r hence:

$$\Sigma_d^r = \bigsqcup_{\rho \in R_d^r} \mathcal{O}_\rho$$

Two representations are in the same G_d -orbits iff they are isomorphic iff they have the same decomposition into indecomposable. So:

$$\Sigma_d^r = \bigsqcup_{\rho \in \mathcal{R}_d^r} \mathcal{O}_\rho$$

We have a stratification in the geometric sense: G_d -orbits parameterized by rank patterns stratify Σ_d^r

Since $\Sigma_d^{r_m} = G_{\text{out}} \cdot \mu^{-1}(M)$ this gives a combinatorial description of the global minima.

The rank patterns are natural geometric invariants observables for DLNs.

Codimension of global minima

For teacher matrix M of rank r_m we have:

$$\text{codim}_{\text{Rep}_d}(\mu^{-1}(M)) = \text{codim}_{\text{Rep}_d}(\Sigma_d^{r_m}) + r_m(d_0 - d_L - r_m) = 2 \text{RLCT}(\mathcal{L}^{-1})(0)$$

In general we only have:

$$\text{RLCT}(\mathcal{L}^{-1})(0) \leq \frac{1}{2} \text{codim}_{\text{Rep}_d}(\mathcal{L}^{-1}(0))$$

DLN are “midly” singular.

Ref: *Lehalleur, Simon Pepin, and Richárd Rimányi. "Geometry of fibers of the multiplication map of deep linear neural networks." arXiv preprint arXiv:2411.19920 (2024).*

Takeaways

Under assumptions of full rank and non degenerate data:

- Critical points of DLNs are either saddles or global minima
- Gauge symmetries means a whole algebraic variety of degenerate minima
- Using quiver representation theory we get a combinatorial description of the global minima
- The Codimension of the global minima is twice the RLCT

Open problems

- Loss landscapes of non linear networks (ReLU)
- Geometric invariants for non linear networks
- Combinatorial description of all critical points (work in progress)

Dynamics

Two layers DLN

$$W := W_L \dots W_1$$

$$W_{>I} := W_L \dots W_{I+1}$$

$$W_{<I} := W_{I-1} \dots W_1$$

$$\Delta(W) := \Sigma_{YX} - W$$

Gradient of the population loss for each layer of a DLN:

$$\dot{W}_I = W_{>I}^\top \nabla_W \mathcal{L}(W) W_{<I}^\top$$

$$\dot{W}_I = W_{>I}^\top \Delta(W) W_{<I}^\top$$

Symmetries induce conserved quantities

The manifold of gradient flow

Gradient flow is constrained on the following hyperbolic manifold:

$$\mathcal{M}_G := \{\theta \in \Omega \mid W_{l+1}^\top W_{l+1} - W_l W_l^\top = G_l\}$$